An algorithm takes 3 seconds for an input size of 3. How long will it take for an input size of 6 if the running time is O(n!)?

a.

6 sec

b.

120 sec

c.

360 sec

d.

240 sec

Feedback

The correct answer is: 360 sec

Question 2

Which of the following growth rates is the most expensive time requirement?

a.

O(n)

b.

O(n!)

c.

O(2^n)

d.

O(n^2)

Feedback

The correct answer is: O(n!)

Question 3

Register r+i of the _____ corresponds to register i of the _____ to simulate a RAM program P by a RASP program.

a.

RASP, RAM

b.

RASP, RASP

c.

RAM, RAM

d.

RAM, RASP

Feedback

The correct answer is: RASP, RAM

Question 4

Uniform Cost Space Complexity for RAM program is _____.

a.

the maximum number of loop for main instruction

b.

the largest integer stored in memory register

c.

the number of variables used

d.

the number of registers used

Feedback

The correct answer is: the number of registers used

Question 5

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

The RASP cost of executing the instruction SUB =i, stored in registers j and j+1 is_____.

a.

l(j+1) + l(c(0)) + l(i)

b.

l(j) + l(j+1) + l(c(0))

c.

l(c(0)) + l(i)

d.

l(j) + l(c(0)) + l(i)

Feedback

The correct answer is: l(j) + l(c(0)) + l(i)

Question 6

The RAM cost of executing the instruction JUMP b, denotes a label, is _____.

a.

l(c(0)) + l(b) + l(c(b))

b.

1

c.

l(c(0))

d.

l(c(0)) + l(b)

Feedback

The correct answer is: 1

Question 7

An algorithm takes 15 seconds for an input size of 500. What will the largest size of input be that can be executed in 120 seconds if the running time is cubic?

a.

2000

b.

4000

c.

6000

d.

1000

Feedback

The correct answer is: 1000

Question 8

Question text

Each RASP instruction requires at most _____ RAM instructions.

a.

6

b.

12

c.

13

d.

2

Feedback

The correct answer is: 13

Question 9

An algorithm takes 3 seconds for an input size of 9. How long will it take for an input size of 12 if the running time is O(2^n)?

a.

36 sec

b.

24 sec

c.

64 sec

d.

16 sec

Feedback

The correct answer is: 24 sec

Question 10

Encoding for RASP instruction MULT =i is_____.

a.

10

b.

9

c.

8

d.

7

Feedback

The correct answer is: 9

Question 11

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

Encoding for RASP instruction JGTZ b, denotes a label, is_____.

a.

16

b.

17

c.

15

d.

18

Feedback

The correct answer is: 16

Question 12

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

The RASP cost of executing the instruction DIV i, stored in registers j and j+1 is _____.

a.

l(c(0)) + l(i) + l(c(i))

b.

l(j) + l(c(0)) + l(i) + l(c(i))

c.

l(j) + l(j+1) + l(c(0)) + l(c(i))

d.

l(j+1) + l(c(0)) + l(i) + l(c(i))

Feedback

The correct answer is: l(j) + l(c(0)) + l(i) + l(c(i))

Question 13

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following functions is ordered by growth rate from smallest to largest?

a.

n log n, 2n^2, 10 log n

b.

10 log n, n log n, 2n^2

c.

2n^2, n log n, 10 log n

d.

n log n, 10 log n, 2n^2

Feedback

The correct answer is: 10 log n, n log n, 2n^2

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following growth rates of a function T is 9 times the run time when we triple the input size (i.e., if T(n) = x, then T(3n) = 9x)?

a.

2^n

b.

3n^2

c.

n^3

d.

n^2

Feedback

The correct answer is: n^2

Question 15

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following growth rates is the slowest function in all?

a.

log5 n

b.

20n

c.

7

d.

n^2/3

Feedback

The correct answer is: 7

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

Question text

When the upper and lower bounds are the same within a constant factor, indicate this by using which of the following asymptotic notation.

a.

None of them

b.

Big-Omega

c.

Big-Oh

d.

Big-Theta

Feedback

The correct answer is: Big-Theta

Question 17

Correct

Mark 1.00 out of 1.00

Flag question

Question text

An algorithm takes 9 seconds for an input size of 200. How large a problem can be solved in 36 seconds if the running time is O(n^2)?

a.

input size of 400

b.

input size of 800

c.

input size of 360

d.

input size of 720

Feedback

The correct answer is: input size of 400

Question 18

Correct

Mark 1.00 out of 1.00

Flag question

Question text

A RAM consists of a read-only input tape, a write-only output tape, another two are_____.

a.

a register and a location counter

b.

a program and a memory

c.

a location counter and an accumulator

d.

none of them

Feedback

The correct answer is: a program and a memory

Question 19

Correct

Mark 1.00 out of 1.00

Flag question

Question text

An algorithm consists of two modules: g1(n) and g2(n). The first module g1(n) is in O(f1(n)) and the second module g2(n) is in O(f2(n)), then g1(n) g2(n) is in _____.

a.

O(max(f1(n), f2(n)))

b.

O(f1(n) f2(n))

c.

O(k f2(n))

d.

none of them

Feedback

The correct answer is: O(f1(n) f2(n))

Question 20

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

Register i+3 of the _____ corresponds to register i of the _____ to simulate a RASP program P by a RAM program.

a.

RAM, RAM

b.

RAM, RASP

c.

RASP, RAM

d.

RASP, RASP

Feedback

The correct answer is: RAM, RASP

Question 21

Correct

Mark 1.00 out of 1.00

Flag question

Question text

An algorithm takes 8 seconds for an input size of 32. What will the largest size of input be that can be executed in 64 seconds if the running time is O(log n)?

a.

2^40

b.

2^8

c.

2^13

d.

2^5

Feedback

The correct answer is: 2^40

Question 22

Correct

Mark 1.00 out of 1.00

Flag question

Question text

An algorithm takes 10 seconds for an input size of 10. How long will it take for an input size of 100 if the running time is O(n log n)?

a.

20 sec

b.

80 sec

c.

200 sec

d.

100 sec

Feedback

The correct answer is: 200 sec

Question 23

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

The RAM cost of executing the instruction STORE *i is _____.

a.

l(c(0)) + l(i) + l(c(i)) + l(c(c(i)))

b.

l(j+1) + l(c(0)) + l(i) + l(c(i))

c.

l(c(0)) + l(i)

d.

l(c(0)) + l(i) + l(c(i))

Feedback

The correct answer is: l(c(0)) + l(i) + l(c(i))

Question 24

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following growth rates of a function T that cubes the run time when we triple the input size (i.e., if T(n) = x, then T(3n) = x^3)?

a.

n^3

b.

2^n^$\log_3$ 2

c.

2^n

d.

2^n^log 3

Feedback

The correct answer is: 2^n

Question 25

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Logarithmic Cost Space Complexity for RAM program is _____.

a.

the number of variables used

b.

the largest integer stored in memory register

c.

the maximum number of loop for main instruction

d.

the number of registers used

Feedback

The correct answer is: the largest integer stored in memory register

Question 26

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following growth rates of a function T is 8 times the run time when we double the input size (i.e., if T(n) = x, then T(2n) = 8x)?

a.

n^3

b.

8n

c.

n^2

d.

2n^3

Feedback

The correct answer is: n^3

Question 27

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

The RAM cost of executing the instruction READ i is _____.

a.

l(c(0)) + l(i)

b.

l(c(0)) + l(i) + l(c(i))

c.

l(input) + l(i) + l(c(i))

d.

l(input) + l(i)

Feedback

The correct answer is: l(input) + l(i)

Question 28

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

Each RAM instruction requires at most _____ RASP instructions.

a.

six

b.

seven

c.

two

d.

three

Feedback

The correct answer is: six

Question 29

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Each RAM instruction consists of two parts: _____.

a.

value and address

b.

Operation code and address

c.

operand and address

d.

operand and operator

Feedback

The correct answer is: Operation code and address

Question 30

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following functions is ordered by growth rate from largest to smallest?

a.

2^4, n log n, n^3, 2^n

b.

n log n, 2^n, n^3, 2^4

c.

2^n, 2^4, n^3, n log n

d.

2^n, n^3, n log n, 2^4

Feedback

The correct answer is: 2^n, n^3, n log n, 2^4

Consider two decision problems Q1, Q2 such that Q2 reduces in polynomial time to 3-SAT and 3-SAT reduces in polynomial time to Q1. Then which one of the following is consistent with the above statement?

a.

Both Q1 and Q2 are in NP hard

b.

Both Q1 and Q2 are in NP

c.

Q2 is in NP, Q1 is NP hard

d.

Q1 is in NP, Q2 is NP hard

Feedback

The correct answer is: Q2 is in NP, Q1 is NP hard

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

Question text

The time complexity of searching on a random sequence using EREW SM SIMD computer is _____.

a.

O(log N)+O(n/N)

b.

O(log N)

c.

O(n/N)

d.

O(log n)

Feedback

The correct answer is: O(log N)+O(n/N)

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Suppose G ≤p H. Which must be true?

a.

Problem H is polynomial-time reducible to problem G.

b.

Problems G and H are polynomial-time equivalent.

c.

All of them

d.

Problem G is polynomial-time reducible to problem H.

Feedback

The correct answer is: Problem G is polynomial-time reducible to problem H.

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which architecture allows multiple processors to execute different instructions on different data simultaneously?

a.

MIMD

b.

SIMD

c.

SISD

d.

MISD

Feedback

The correct answer is: MIMD

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Suppose problem C is NP-complete, problem D is in class NP, and C is polynomial-time reducible to D. Which must be true?

a.

both of them

b.

none of them

c.

problem D is in class P

d.

problem D is NP–complete

Feedback

The correct answer is: problem D is NP–complete

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Let S = {7, 14, 6, 8, 11, 12, 16, 13} and x= 9. Which of the following is TRUE for searching closet element on a tree-connected SIMD computer?

a.

(1,8)

b.

(4,1)

c.

(8,1)

d.

(1,4)

Feedback

The correct answer is: (4,1)

Question 7

Correct

Mark 1.00 out of 1.00

Flag question

Question text

CUBE MATRIX MULTIPLICATION of two 4×4 matrices require _____ bits for representing each processor.

a.

6

b.

8

c.

9

d.

4

Feedback

The correct answer is: 6

Question 8

Correct

Mark 1.00 out of 1.00

Flag question

Question text

The _____ of the element on a tree is searched by adding 1 to the sum of its two inputs by root before its produce.

a.

Position

b.

Rank

c.

All of them

d.

Count

Feedback

The correct answer is: Rank

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Language L1 is polynomial time reducible to language L2. Language L3 is polynomial time reducible to L2, which in turn is polynomial time reducible to language L4. Which of the following is True?

a.

If L2 ∈ P, then L4 ∈ P

b.

L1 ∈ P, if and only if L3 ∈ P

c.

If L4 ∈ P, then L1 ∈ P and L3 ∈ P

d.

If L1 ∈ P or L3 ∈ P, then L2 ∈ P

Feedback

The correct answer is: If L4 ∈ P, then L1 ∈ P and L3 ∈ P

Question 10

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Let X be a problem that belongs to the class NP. Then which one of the following is TRUE?

a.

There is no polynomial time algorithm for X.

b.

If X is NP-hard, then it is NP-complete.

c.

If X can be solved deterministically in polynomial time, then P = NP.

d.

X may be undecidable.

Feedback

The correct answer is: If X is NP-hard, then it is NP-complete.

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

Question text

For problems X and Y, Y is NP-Complete and X reduces to Y in polynomial time. Which of the following is TRUE?

a.

If X can be solved in polynomial time, then so can Y

b.

X is NP-hard

c.

X is NP-complete

d.

X is in NP, but not necessarily NP-complete

Feedback

The correct answer is: X is in NP, but not necessarily NP-complete

Question 12

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

Let S= {28, 11, 6, 8, 11, 15, 17, 23} and the search key=11. Which of the following is TRUE for searching on a tree-connected SIMD computer?

a.

The root of the tree produces (1, 5) to give the position of search key.

b.

The root produces 3 to count the number of records whose s field equal x.

c.

The root produces 3 for rank.

d.

None of them.

Feedback

The correct answer is: The root produces 3 for rank.

Question 13

Correct

Mark 1.00 out of 1.00

Flag question

Question text

In MESH MATRIX MULTIPLICATION of A [m×n] and B [n×k], Elements a_{m1} and b_{1k} take _____ steps from the beginning of the computation to reach P(m, k).

a.

m+k+n-3

b.

m+k+n-2

c.

m+k+n

d.

m+k+n-1

Feedback

The correct answer is: m+k+n-2

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Among parallel algorithms for searching on the sorted sequence, _____ SEARCHING has a time complexity of O(log n) to search the key.

a.

CRCW

b.

CREW

c.

EREW

d.

ERCW

Feedback

The correct answer is: EREW

Question 15

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

Let S = {35, 34, 56, 28, 35, 27, 29, 27} and the search key x=27. Which of the following is false for searching on a tree-connected SIMD computer?

a.

The root produces 2 to count the number of records whose s field equals x.

b.

All of them.

c.

The root of the tree produces (1,6) to give the position of search key.

d.

The root produces (3,2) to give the closest value of x.

Feedback

The correct answer is: The root produces (3,2) to give the closest value of x.

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

Question text

A problem in NP is NP-complete if _____.

a.

it can be reduced to the 3-SAT problem in polynomial time

b.

some problem in NP can be reduced to it in polynomial time

c.

the 3-SAT problem can be reduced to it in polynomial time

d.

it can be reduced to any other problem in NP in polynomial time

Feedback

The correct answer is: the 3-SAT problem can be reduced to it in polynomial time

Question 17

Incorrect

Mark 0.00 out of 1.00

Flag question

Question text

If D is NP-complete and D is in P, then _____.

a.

All of them

b.

P = NP

c.

NP-complete = NP

d.

P = NP-complete

Feedback

The correct answer is: All of them

Question 18

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following is true about NP-Complete and NP-Hard problems.

a.

If we want to prove that a problem X is NP-Hard, we take a known NP-Hard problem Y and reduce Y to X

b.

The first problem that was proved as NP-complete was the circuit satisfiability problem

c.

NP-complete is a subset of NP Hard

d.

All of them

Feedback

The correct answer is: All of them

Question 19

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following statements are TRUE?

a.

If a problem is in NP, it is solvable in polynomial time.

b.

2-SAT is NP-Complete.

c.

If P=NP, then every NP problem has a polynomial-time algorithm.

d.

none of them

Feedback

The correct answer is: If P=NP, then every NP problem has a polynomial-time algorithm.

Question 20

Correct

Mark 1.00 out of 1.00

Flag question

Question text

In n x n MESH TRANSPOSE, the longest path traversed by element a_{ij} to P(j, i) consists of _____ steps.

a.

n-1

b.

2n-1

c.

n-2

d.

2n-2

Feedback

The correct answer is: 2n-2

Question 21

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Matrix-by-vector multiplication of A [m x n] by B[n x 1] requires _____ processors on a LINEAR array.

a.

m

b.

m×n

c.

n

d.

n^2

Feedback

The correct answer is: m

Question 22

Correct

Mark 1.00 out of 1.00

Flag question

Question text

The problem 3-SAT and 2-SAT are _____.

a.

both NP complete

b.

undecidable and NP-complete respectively

c.

NP-complete and in P respectively

d.

both in P

Feedback

The correct answer is: NP-complete and in P respectively

Question 23

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following statements are TRUE? (1)Every problem in P is also in NP. (2)If a problem is NP-Complete, then it is NP-Hard. (3)If a problem is NP-Hard, then it must belong to NP.

a.

1 and 2

b.

2 and 3

c.

1 and 3

d.

1, 2 and 3

Feedback

The correct answer is: 1 and 2

Question 24

Correct

Mark 1.00 out of 1.00

Flag question

Question text

In CREW searching, the value of x must be made known to all processors in _____ time.

a.

O(n/N)

b.

O(logN)

c.

O(n/N)

d.

O(1)

Feedback

The correct answer is: O(1)

Question 25

Correct

Mark 1.00 out of 1.00

Flag question

Question text

In LINEAR MV MULTIPLICATION of A[m×n] by U[n×1], element a_{1n} takes _____ steps to reach P1.

a.

m+k+n-3

b.

n×1

c.

m+n

d.

m+n-1

Feedback

The correct answer is: m+n-1

Question 26

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Which of the following is true for a tree-connected SIMD computer which can search a file of n records?

a.

All of them.

b.

There are n leaves in total.

c.

The root is in charge of receiving input from the outside world

d.

Intermediate node can receive one input from its parent and two inputs from its children.

Feedback

The correct answer is: All of them.

Question 27

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Let S be an NP-complete problem and Q and R be two other problems not known to be in NP. R is polynomial time reducible to S and S is polynomial-time reducible to Q. Which one of the following statements is true?

a.

Q is NP-hard

b.

R is NP-hard

c.

R is NP-complete

d.

Q is NP-complete

Feedback

The correct answer is: Q is NP-hard

Question 28

Correct

Mark 1.00 out of 1.00

Flag question

Question text

In TREE MV MULTIPLICATION of A[5×4] by U[4×1], there are _____ intermediate processors.

a.

1

b.

3

c.

2

d.

4

Feedback

The correct answer is: 2

Question 29

Correct

Mark 1.00 out of 1.00

Flag question

Question text

In SHUFFLE TRANSPOSE of matrix A [8×8], the element a_{32} is initially stored in the processor_____.

a.

P16

b.

P18

c.

P17

d.

P15

Feedback

The correct answer is: P17

Question 30

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Let S = {25, 24, 46, 28, 26, 27, 29, 27} and the search key x=24. How many processors are needed for searching x on a mesh-connected SIMD computer?

a.

15

b.

16

c.

8

d.

6

Feedback

The correct answer is: 8